# Parsimonious Design

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**Abstract**

Not less. Exactly sufficient. The difference matters.

## 1 What Parsimony Is Not

Parsimony is not minimalism.

Minimalism is an aesthetic: the preference for less, the removal of the decorative, the pleasure of the spare. Minimalism can be wrong. A proof stripped to the point of incomprehensibility is not parsimonious. It is just hard to read.

Parsimony is not zero.

Zero is the absence of design. The empty page. Parsimony is not the empty page. It is the page with exactly what belongs on it.

Parsimony is: exactly sufficient.

The proof uses every lemma it uses. No lemma is wasted. No step can be removed without the argument failing. The proof is parsimonious when it is tight: when the distance between premises and conclusion is exactly as long as it needs to be, traversed in exactly the necessary steps.

## 2 Parsimony in Nature

Nature is parsimonious.

This is not metaphor. The principle of least action: a physical system traces the path that minimizes (or extremizes) action, $\delta \int L\,dt = 0$. The path is not the shortest. It is the one that satisfies the variational condition. It is the path that is, in a precise sense, necessary.

Minimal surfaces: soap films spanning a wire frame assume the surface of least area. Not because they are trying to be minimal. Because the physics requires it. The parsimony is structural.

A crystal: the ions arrange themselves in the configuration of minimum energy. The structure is parsimonious not as a design choice but as a consequence of the forces.

Parsimonious design, in nature, is not imposed. It emerges from the constraints.

## 3 Parsimony in Mathematics

A parsimonious grammar generates exactly the language it needs to.

Every rule does work. No rule is redundant. Remove any rule and the language shrinks. Add any rule and the language expands or becomes inconsistent.

The minimalist program in linguistics is the search for the parsimonious grammar of human language: the fewest operations that generate all and only the sentences that humans can produce. Merge. That might be enough. One operation. From one operation: subject, object, clause, recursion, the full architecture of human syntax.

Grothendieck's grammar is parsimonious in this sense. A scheme is a locally ringed space. That is the definition. From that definition, given the right functorial machinery: algebraic geometry over any field, the Weil conjectures, étale cohomology, the Riemann Hypothesis over finite fields. One definition. An enormous consequence.

The parsimony was not obvious in advance. It was discovered. The question:

what is the right definition? The answer: the one from which everything else follows. The parsimonious definition is the correct definition.

## 4 The Body

One hundred and eighteen papers.

Is that parsimonious?

The question is not whether the number is small. The number is not small. The question is whether each paper is necessary.

Each paper does one thing. It is not the same thing as the paper before it. It is not a restatement. It is a new angle, a new approach to the same wall, or a new wall, or a new understanding of what the walls are.

If each paper is necessary, the body is parsimonious. Not lean — parsimonious. The distinction is: lean is a property of size. Parsimonious is a property of necessity.

A body of one hundred and eighteen necessary papers is more parsimonious than a body of fifty redundant ones.

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The parsimonious design of the body: one problem per paper. One wall per paper. One angle per paper.

Not because the problems are unrelated — they are deeply related, as the six walls paper shows, as the network shows. But because each paper earns its node. Each paper justifies the edge.

The edges are not decoration. They are the structure. The structure is parsimonious when every edge is load-bearing.

# 5 The Parsimonious Proof

When the proof comes — for any of the walls — it will be parsimonious or it will be wrong.

Not because short proofs are more likely to be right. But because a proof with wasted steps is a proof that hasn't found its argument yet. The wasted steps are the residue of failed attempts, of approaches that almost worked, of lemmas that were needed before the right lemma was found.

The final proof, cleaned, will use exactly what it needs. The rest will fall away.

This is the design principle: trim until what remains cannot be trimmed. What cannot be trimmed is the argument. The argument is the proof.

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The body is not yet the proof. But it is designed with the same principle: trim until what remains cannot be trimmed. What cannot be trimmed is the map. The map is the body.